

5.To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_INPUT 1024
#define MAX_ARGS 100

int main()
{
    char input[MAX_INPUT];
    char *args[MAX_ARGS];

    printf("===== Mini Shell =====\n");
    printf("Type 'exit' to quit.\n\n");

    while (1)
    {
        printf("MiniShell> ");

        if (fgets(input, sizeof(input), stdin) == NULL)
            break;

        input[strcspn(input, "\n")] = '\0';

        if (strlen(input) == 0)
            continue;

        if (strcmp(input, "exit") == 0)
            break;

        /* Replace escaped spaces (\ ) with a temporary character */
        for (int i = 0; input[i] != '\0'; i++)
        {
            if (input[i] == '\\' && input[i + 1] == ' ')
            {
                input[i] = 1;          // Temporary marker
                memmove(&input[i + 1], &input[i + 2], strlen(&input[i + 2]) + 1);
            }
        }

        int argc = 0;
        char *token = strtok(input, " ");
```

```

prajwal@PRAJWAL:~/os/OSSkill$ ./skill5
===== Mini Shell =====
Type 'exit' to quit.

MiniShell> hello world
Execution Failed: No such file or directory
MiniShell> ls -l
total 116
-rw-r--r-- 1 prajwal prajwal  583 Aug 17 09:24 skill1.c
-rwxr-xr-x 1 prajwal prajwal 16392 Sep 18 08:07 skill4
-rw-r--r-- 1 prajwal prajwal  2748 Sep 18 08:07 skill4.c
-rwxr-xr-x 1 prajwal prajwal 16560 Sep 28 04:42 skill5
-rw-r--r-- 1 prajwal prajwal  1715 Sep 28 04:41 skill5.c
-rwxr-xr-x 1 prajwal prajwal 16216 Aug 24 06:43 skillq1
-rw-r--r-- 1 prajwal prajwal   815 Aug 24 06:43 skillq1.c
-rwxr-xr-x 1 prajwal prajwal 16424 Aug 24 06:46 skillq2
-rw-r--r-- 1 prajwal prajwal  2266 Aug 24 06:46 skillq2.c
-rwxr-xr-x 1 prajwal prajwal 16264 Aug 24 06:49 skillq3
-rw-r--r-- 1 prajwal prajwal  1669 Aug 24 06:48 skillq3.c
MiniShell> |

```

6.To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching.

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_INPUT 1024
#define MAX_ARGS 64

// Simple tokenizer: splits input by spaces
void parse_input(char *input, char **args) {
    char *token = strtok(input, " \t\n");
    int i = 0;
    while (token != NULL && i < MAX_ARGS - 1) {
        args[i++] = token;
        token = strtok(NULL, " \t\n");
    }
    args[i] = NULL; // Null terminate
}

int main() {
    char input[MAX_INPUT];
    char *args[MAX_ARGS];
    pid_t pid;

    while (1) {
        printf("myshell> "); // prompt
        if (fgets(input, MAX_INPUT, stdin) == NULL) {
            break; // exit on Ctrl+D
        }

        parse_input(input, args);

        if (args[0] == NULL) continue; // empty input

        if (strcmp(args[0], "exit") == 0) {
            break; // quit shell
        }

        pid = fork();
        if (pid == 0) {
            // Child process
            if (execvp(args[0], args) == -1) {
                perror("execvp failed");
            }
        }
    }
}

```

```
prajwal@PRAJWAL:~/os/OSSkill$ ./skill6
```

```
myshell> ls -l
```

```
total 140
```

```

-rw-r--r-- 1 prajwal prajwal  583 Aug 17 09:24 skill1.c
-rwxr-xr-x 1 prajwal prajwal 16392 Sep 18 08:07 skill4
-rw-r--r-- 1 prajwal prajwal  2748 Sep 18 08:07 skill4.c
-rwxr-xr-x 1 prajwal prajwal 16560 Sep 28 04:42 skill5
-rw-r--r-- 1 prajwal prajwal  1715 Sep 28 04:41 skill5.c
-rwxr-xr-x 1 prajwal prajwal 16424 Sep 28 05:06 skill6
-rw-r--r-- 1 prajwal prajwal  1253 Sep 28 05:06 skill6.c
-rwxr-xr-x 1 prajwal prajwal 16216 Aug 24 06:43 skillq1
-rw-r--r-- 1 prajwal prajwal   815 Aug 24 06:43 skillq1.c
-rwxr-xr-x 1 prajwal prajwal 16424 Aug 24 06:46 skillq2
-rw-r--r-- 1 prajwal prajwal   2266 Aug 24 06:46 skillq2.c
-rwxr-xr-x 1 prajwal prajwal 16264 Aug 24 06:49 skillq3
-rw-r--r-- 1 prajwal prajwal   1669 Aug 24 06:48 skillq3.c
myshell> |

```

7. Process Synchronization and Child Process Monitoring Using waitpid().

1. To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>
#include <sys/stat.h>

#define MAX_INPUT 1024
#define MAX_ARGS 64

// Parse input into arguments
void parse_input(char *input, char **args) {
    char *token = strtok(input, " \\t\\n");
    int i = 0;
    while (token != NULL && i < MAX_ARGS - 1) {
        args[i++] = token;
        token = strtok(NULL, " \\t\\n");
    }
    args[i] = NULL;
}

// Check if command exists in PATH
char *resolve_command(char *cmd) {
    char *path = getenv("PATH");
    char *copy = strdup(path);
    char *dir = strtok(copy, ":");
    static char fullpath[1024];
    struct stat sb;

    while (dir != NULL) {
        snprintf(fullpath, sizeof(fullpath), "%s/%s", dir, cmd);
        if (stat(fullpath, &sb) == 0 && sb.st_mode & S_IXUSR) {
            free(copy);
            return fullpath; // executable found
        }
        dir = strtok(NULL, ":");
    }
    free(copy);
    return NULL; // not found
}

int main() {
    char input[MAX_INPUT];
    char *args[MAX_ARGS];
    pid_t pid;
```

```

prajwal@PRAJWAL:~/os/OSSkill$ nano skill7.c
prajwal@PRAJWAL:~/os/OSSkill$ gcc skill7.c -o skill7
prajwal@PRAJWAL:~/os/OSSkill$ ./skill7
myshell> ls -l
total 164
-rw-r--r-- 1 prajwal prajwal  583 Aug 17 09:24 skill1.c
-rwxr-xr-x 1 prajwal prajwal 16392 Sep 18 08:07 skill4
-rw-r--r-- 1 prajwal prajwal  2748 Sep 18 08:07 skill4.c
-rwxr-xr-x 1 prajwal prajwal 16560 Sep 28 04:42 skill5
-rw-r--r-- 1 prajwal prajwal  1715 Sep 28 04:41 skill5.c
-rwxr-xr-x 1 prajwal prajwal 16424 Sep 28 05:06 skill6
-rw-r--r-- 1 prajwal prajwal  1253 Sep 28 05:06 skill6.c
-rwxr-xr-x 1 prajwal prajwal 16696 Sep 28 05:10 skill7
-rw-r--r-- 1 prajwal prajwal  2244 Sep 28 05:10 skill7.c
-rwxr-xr-x 1 prajwal prajwal 16216 Aug 24 06:43 skillq1
-rw-r--r-- 1 prajwal prajwal   815 Aug 24 06:43 skillq1.c
-rwxr-xr-x 1 prajwal prajwal 16424 Aug 24 06:46 skillq2
-rw-r--r-- 1 prajwal prajwal  2266 Aug 24 06:46 skillq2.c
-rwxr-xr-x 1 prajwal prajwal 16264 Aug 24 06:49 skillq3
-rw-r--r-- 1 prajwal prajwal  1669 Aug 24 06:48 skillq3.c
Child exited with status 0

```

8.To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX 1000

void expand_variables(char *input)
{
    char result[MAX] = "";
    char var[100];
    int i = 0;

    while (input[i] != '\0')
    {
        if (input[i] == '$')
        {
            i++;

            int j = 0;

            // ${VARIABLE} format
            if (input[i] == '{')
            {
                i++;

                while (input[i] != '}' && input[i] != '\0')
                    var[j++] = input[i++];

                if (input[i] == '}')
                    i++;
            }
            // $VARIABLE format
            else
            {
                while ((input[i] >= 'A' && input[i] <= 'Z') ||
                       (input[i] >= 'a' && input[i] <= 'z') ||
                       (input[i] >= '0' && input[i] <= '9') ||
                       input[i] == '_')
                {
                    var[j++] = input[i++];
                }
            }
        }

        result[i] = var[j];
        i++;
    }

    result[i] = '\0';
}

```

```

prajwal@PRAJWAL:~/os/OSSkill$ nano skill8.c
prajwal@PRAJWAL:~/os/OSSkill$ gcc skill8.c -o skill8
prajwal@PRAJWAL:~/os/OSSkill$ ./skill8
Variable Expansion Shell
Enter a command: My home is $HOME

Before expansion:
My home is $HOME

After expansion:
My home is /home/prajwal

```